

MAC21 — 4-Mark Bank, Every Single Step

Same 45 questions, but nothing skipped. **Every derivative is computed in front of you, every integral is named, every line of algebra is shown.** No step assumes you already know calculus.

NO CALCULUS = pure arithmetic, you can do these today

NEEDS CALCULUS = uses a derivative or integral, all of which are in the toolkit below.

Good news first: Units IV and V are almost entirely arithmetic. Difference tables, Gauss-Seidel, Lagrange, trapezoidal, determinants — that is 15 of the 45 questions with *no calculus at all*. If maths is fighting you, start there.

THE TOOLKIT — every rule these 45 questions actually use

Differentiation — "rate of change". Written d/dx or f'

RULE	WHAT IT DOES	EXAMPLE
$d/dx (x^n) = n \cdot x^{n-1}$	Bring the power down in front, then reduce the power by 1.	$x^3 \rightarrow 3x^2 \cdot x \rightarrow 1 \cdot 5 \rightarrow 0$
$d/dx (e^x) = e^x$	Unchanged. e is a number ≈ 2.718 .	$e_x \rightarrow e_x$
$d/dx (e^{a \cdot x}) = a \cdot e^{a \cdot x}$	Copy it, multiply by whatever multiplies x .	$e^2_x \rightarrow 2e^2_x \cdot e^{-2}_x \rightarrow -2e^{-2}_x$
$d/dx (a^x) = a^x \cdot \log a$	Copy it, multiply by $\log a$ (a constant).	$2_x \rightarrow 2_x \log 2$
$d/dx (\sin x) = \cos x$ $d/dx (\cos x) = -\sin x$	They swap. Cosine picks up a minus sign.	$\sin x \rightarrow \cos x \rightarrow -\sin x \rightarrow -\cos x$ (cycles every 4)
$d/dx (\log x) = 1/x$		$\log x \rightarrow 1/x$
$d/dx (\tan^{-1}x) = 1/(1+x^2)$ $d/dx (\sin^{-1}x) = 1/\sqrt{1-x^2}$	Just memorise these two.	—
Product rule $(u \cdot v)' = u'v + u \cdot v'$	Two things multiplied: differentiate the first times the second, plus the first times the derivative of the second.	$x e_x \rightarrow (1)(e_x) + (x)(e_x) = e_x(1+x)$
Chain rule $d/dx f(g) = f'(g) \cdot g'$	Something inside something: differentiate the outside, then multiply by the derivative of the inside.	$e^{-2 \cdot t} \rightarrow e^{-2 \cdot t} \cdot (-2)$
Partial derivative $\partial f / \partial x$, written f_x	Two variables. To get f_x , treat y as if it were a plain number and differentiate normally. For f_y , treat x as a number.	$f = x^2y$: $f_x = 2xy$ (y is "a number") $f_y = x^2$ (x^2 is "a number")

Integration — the reverse of differentiation. Written $\int \dots dx$

RULE	WHAT IT DOES	EXAMPLE
$\int x^n dx = x^{n+1}/(n+1)$	Raise the power by 1, divide by the new power. (Opposite of differentiating.)	$\int x^2 dx = x^3/3 \cdot \int 1 dx = x$
$\int e^{a \cdot x} dx = e^{a \cdot x}/a$	Copy it, <i>divide</i> by whatever multiplies x .	$\int e^2_x dx = e^2_x/2$
$\int (1/x) dx = \log x$	The one exception to the power rule.	$\int dx/x = \log x$
$\int \tan \theta d\theta = -\log(\cos \theta)$ $\int \cot \theta d\theta = \log(\sin \theta)$	Needed only for orthogonal-trajectory questions.	—

Two facts about logs you will need

$\log A + \log B = \log(AB)$ · $n \cdot \log A = \log(A^n)$ · if $\log P = \log Q$ then $P = Q$ · $\log(e^k) = k$

UNIT I — Q1(b) or Q2(b)

1 Show $f(x, y) = xy(1-x-y)$ is maximum at $(1/3, 1/3)$ **3x** NEEDS CALCULUS

Step 1 Multiply it out first — much easier to differentiate.

$$f = xy(1 - x - y) = xy - x^2y - xy^2$$

Step 2 Find f_x . **Treat y as a plain number.**

• $xy \rightarrow y$ is a number multiplying x , so this differentiates to y

• $-x^2y \rightarrow -y \cdot (2x) = -2xy$

• $-xy^2 \rightarrow y^2$ is a number, so this gives $-y^2$

$$f_x = y - 2xy - y^2$$

Step 3 Find f_y . **Treat x as a plain number.**

$$f_y = x - x^2 - 2xy$$

Step 4 Set both to zero and check the given point works.

Put $x = 1/3$, $y = 1/3$ into f_x :

$$1/3 - 2(1/3)(1/3) - (1/3)^2 = 1/3 - 2/9 - 1/9 = 3/9 - 2/9 - 1/9 = 0 \quad \checkmark$$

By symmetry $f_y = 0$ too. So $(1/3, 1/3)$ is a stationary point.

Step 5 Differentiate a *second* time to get r , s , t .

• $r = f_{xx}$ = differentiate $f_x = y - 2xy - y^2$ w.r.t. x again $\rightarrow$ only $-2xy$ has an $x \rightarrow r = -2y$

• $s = f_{xy}$ = differentiate f_x w.r.t. $y \rightarrow s = 1 - 2x - 2y$

• $t = f_{yy}$ = differentiate $f_y = x - x^2 - 2xy$ w.r.t. $y \rightarrow t = -2x$

Step 6 Put $(1/3, 1/3)$ in — now it is only arithmetic.

$$r = -2/3 \quad s = 1 - 2/3 - 2/3 = -1/3 \quad t = -2/3$$

$$rt - s^2 = (-2/3)(-2/3) - (-1/3)^2 = 4/9 - 1/9 = 3/9 = 1/3$$

Step 7 Apply the test.

$rt - s^2 = 1/3 > 0$ and $r = -2/3 < 0 \rightarrow$ **MAXIMUM**

$$\text{Value: } f = (1/3)(1/3)(1 - 1/3 - 1/3) = (1/9)(1/3) = \mathbf{1/27}$$

2 Examine $x^3 + y^3 - 3axy$ for extreme values **1x** NEEDS CALCULUS

Step 1 f_x : treat y as a number. $x^3 \rightarrow 3x^2$; $y^3 \rightarrow 0$ (no x in it); $-3axy \rightarrow -3ay$.

$$f_x = 3x^2 - 3ay \quad \text{and similarly} \quad f_y = 3y^2 - 3ax$$

Step 2 Set both zero: $x^2 = ay$ and $y^2 = ax$.

From the first, $y = x^2/a$. Put that into the second: $(x^2/a)^2 = ax \rightarrow x^4/a^2 = ax \rightarrow x^4 = a^3x \rightarrow x(x^3 - a^3) = 0$

So $x = 0$ or $x = a$. Stationary points: $(0, 0)$ and (a, a) .

Step 3 Second derivatives: $r = f_{xx} = 6x$, $s = f_{xy} = -3a$, $t = f_{yy} = 6y$

$$rt - s^2 = 36xy - 9a^2$$

Step 4 Test each point.

At $(0, 0)$: $rt - s^2 = 0 - 9a^2 = -9a^2 < 0 \rightarrow$ **saddle point**

At (a, a) : $rt - s^2 = 36a^2 - 9a^2 = 27a^2 > 0$, and $r = 6a$

$\rightarrow$ if $a > 0$, $r > 0$ so **minimum**; if $a < 0$, **maximum**

$$\text{Value there: } a^3 + a^3 - 3a \cdot a \cdot a = \mathbf{-a^3}$$

3 Expand a_x in powers of x , first three terms **1x** NEEDS CALCULUS

Formula Maclaurin: $f(x) = f(0) + x f'(0) + (x^2/2!) f''(0) + \dots$ ($2! = 2$)

Step 1 $f = a_x \rightarrow$ at $x=0$: $a^0 = 1$

Step 2 $f' = a_x \cdot \log a \rightarrow$ at 0: $1 \cdot \log a = \log a$

Step 3 f'' : differentiate $a_x \log a$. $\log a$ is a constant, so it just rides along: $f'' = a_x (\log a)^2 \rightarrow$ at 0: $(\log a)^2$

Step 4 Substitute: $a_x = \mathbf{1 + x \log a + x^2(\log a)^2/2 + \dots}$

4 Expand $\sin^{-1}x$ in powers of x 1x NEEDS CALCULUS

Trick Differentiating $\sin^{-1}x$ repeatedly is horrible. Instead: differentiate *once*, expand the simple thing you get, then integrate back.

Step 1 $d/dx(\sin^{-1}x) = 1/\sqrt{1-x^2} = (1-x^2)^{-\frac{1}{2}}$

Step 2 Binomial expansion of $(1-u)^{-\frac{1}{2}}$ is $1 + u/2 + 3u^2/8 + \dots$. Put $u = x^2$:
 $(1-x^2)^{-\frac{1}{2}} = 1 + x^2/2 + 3x^4/8 + \dots$

Step 3 Integrate term by term (raise the power by 1, divide by the new power):
 $\int 1 dx = x \quad \cdot \quad \int (x^2/2) dx = x^3/6 \quad \cdot \quad \int (3x^4/8) dx = 3x^5/40$

Answer $\sin^{-1}x = x + x^3/6 + 3x^5/40 + \dots$

The constant is 0 because $\sin^{-1}(0) = 0$. All even powers are missing — say so, it is worth a mark.

5 Expand $\tan^{-1}x$ in powers of $(x-1)$ to 4th degree 1x NEEDS CALCULUS

Formula Taylor about $a=1$: $f(x) = f(1) + (x-1)f'(1) + (x-1)^2 f''(1)/2! + (x-1)^3 f'''(1)/3! + (x-1)^4 f^{(4)}(1)/4!$
 $2! = 2$, $3! = 6$, $4! = 24$

Step 1 $f = \tan^{-1}x \rightarrow f(1) = \tan^{-1}(1) = \pi/4$

Step 2 $f' = 1/(1+x^2) \rightarrow f'(1) = 1/2$

Step 3 $f'' = -2x/(1+x^2)^2 \rightarrow f''(1) = -2/4 = -1/2$

Step 4 $f''' = (6x^2-2)/(1+x^2)^3 \rightarrow f'''(1) = 4/8 = 1/2$

Step 5 $f^{(4)}(1) = 0$ (it works out to zero at $x=1$)

Step 6 Substitute, dividing by the factorials:

$$\pi/4 + (x-1)(1/2) + (x-1)^2(-1/2)/2 + (x-1)^3(1/2)/6 + 0$$

Answer $\tan^{-1}x = \pi/4 + (x-1)/2 - (x-1)^2/4 + (x-1)^3/12 + 0 \cdot (x-1)^4$

6 Newton-Raphson system, ONE iteration, from $(0.8, 0.4)$ 1x easy calculus only

Setup $x^2+y^2 = x$ and $x^2-y^2 = y$. Move everything to one side:

$$f = x^2 + y^2 - x \quad g = x^2 - y^2 - y$$

Step 1 Four partial derivatives (all just power rule):

$$f_x = 2x - 1 \quad f_y = 2y \quad g_x = 2x \quad g_y = -2y - 1$$

Step 2 Evaluate everything at $(0.8, 0.4)$ — pure arithmetic from here.

$$f = 0.64 + 0.16 - 0.8 = 0.0000 \quad (\text{zero — that is fine, not an error})$$

$$g = 0.64 - 0.16 - 0.4 = 0.0800$$

$$f_x = 0.6, \quad f_y = 0.8, \quad g_x = 1.6, \quad g_y = -1.8$$

Step 3 Solve the two linear equations for the corrections h, k :

$$0.6h + 0.8k = -0.0000$$

$$1.6h - 1.8k = -0.0800$$

From the first, $h = -(0.8/0.6)k = -1.3333k$. Substitute:

$$1.6(-1.3333k) - 1.8k = -0.08 \rightarrow -2.1333k - 1.8k = -0.08 \rightarrow -3.9333k = -0.08$$

$$k = 0.020339, \quad h = -1.3333(0.020339) = -0.027119$$

Step 4 Update: $x_1 = 0.8 - 0.027119$, $y_1 = 0.4 + 0.020339$

$$x_1 = 0.772881, \quad y_1 = 0.420339$$

7 Newton-Raphson system from $(3.5, -1.8)$, one iteration 1x

Setup $f = x^2 + y - 11$, $g = y^2 + x - 7$

Step 1 $f_x = 2x$, $f_y = 1$, $g_x = 1$, $g_y = 2y$

Step 2 At $(3.5, -1.8)$:

$$f = 12.25 - 1.8 - 11 = -0.5500 \quad \cdot \quad g = 3.24 + 3.5 - 7 = -0.2600$$

$$f_x = 7, \quad f_y = 1, \quad g_x = 1, \quad g_y = -3.6$$

Step 3 $7h + k = 0.55$ and $h - 3.6k = 0.26$

From the first, $k = 0.55 - 7h$. Substitute: $h - 3.6(0.55 - 7h) = 0.26$

$$h - 1.98 + 25.2h = 0.26 \rightarrow 26.2h = 2.24 \rightarrow h = 0.085496$$

$$k = 0.55 - 7(0.085496) = -0.048473$$

Answer $x_1 = 3.585496$, $y_1 = -1.848473$

8 Negative root of $x e^x - \sin x = 0$ to three decimals 2x NEEDS CALCULUS

Step 1 — find where the root is. Put in numbers until the sign flips:

$$f(-2.9) = (-2.9)(0.0550) - \sin(-2.9) = -0.1596 + 0.2392 = +0.0796$$

$$f(-3.0) = (-3.0)(0.0498) - \sin(-3.0) = -0.1494 + 0.1411 = -0.0083$$

Sign changes $\rightarrow$ root lies between -3.0 and -2.9 . Start at $x_0 = -3$.

Do not start near 0 — $x=0$ is also a root and the method will slide into it.

Step 2 — the derivative. $x e^x$ is a product, so use the product rule:

$$d/dx(x e^x) = (1)(e^x) + (x)(e^x) = e^x(1+x); \text{ and } d/dx(-\sin x) = -\cos x$$

$$f'(x) = e^x(1+x) - \cos x$$

Step 3 — iterate $x_{-1} = x - f(x)/f'(x)$.

$$\text{At } x = -3: f = -0.008240; f' = 0.0498(-2) - (-0.98999) = -0.0996 + 0.98999 = 0.89039$$

$$x_1 = -3 - (-0.008240/0.89039) = -3 + 0.009255 = -2.990745$$

Repeat once: $x_2 = -2.990735$ — the first three decimals have stopped moving.

Answer **root = -2.991**

UNIT II — Q3(b) or Q4(b)

11 Bungee jumper — Euler's method 2x NO CALCULUS

Given $dv/dt = g - (c/m)v^2$, $g = 9.8$, $c = 0.25$, $m = 68.1$, $v(0) = 0$, $h = 1$ second.

Step 1 Work out the constant once: $c/m = 0.25/68.1 = 0.003671$.

So $f(t, v) = 9.8 - 0.003671 v^2$. Keep 6 figures — rounding to 0.004 changes the answer.

Step 2 Euler's rule: $v_{-1} = v_n + h \cdot f$. With $h=1$ that is just $v + f$.

T	V	F = $9.8 - 0.003671 V^2$	NEW V
0	0	$9.8 - 0 = 9.8000$	9.8000
1	9.8000	$9.8 - 0.003671(96.04) = 9.8 - 0.3526 = 9.4474$	19.2474
2	19.2474	$9.8 - 0.003671(370.46) = 9.8 - 1.3600 = 8.4400$	27.6874

Answer **v(3) = 27.69 m/s**

12 Euler: $dy/dx - 2y = 3e^x$, $y(0)=0$, find $y(0.2)$, $h=0.1$ 2x NO CALCULUS

Step 1 Rearrange so dy/dx is alone: $dy/dx = 2y + 3e^x$. So $f(x, y) = 2y + 3e^x$.

Step 2 $y_{-1} = y_n + h \cdot f(x_n, y_n)$ with $h = 0.1$.

STEP	X	Y	F = $2Y + 3e^x$	Y + 0.1F
1	0	0	$0 + 3(1) = 3.000000$	0.300000
2	0.1	0.300000	$0.6 + 3(1.105171) = 3.915513$	0.691551

Answer **y(0.2) = 0.691551** ($e^{0.1} = 1.105171$)

Variant Same DE with $y(1)=0$, $h=0.05$, to $x=1.1$:

$$y(1.05) = 0 + 0.05(0 + 3e^1) = 0.05(8.154845) = 0.407742$$

$$y(1.10) = 0.407742 + 0.05(0.815484 + 3 \times 2.857651) = \mathbf{0.877164}$$

14 Taylor's series method: $dy/dt = e^t - 2y$, $y(0)=1$, at $t=0.1$ 2x NEEDS CALCULUS

The formula $y(t) = y_0 + t \cdot y'_0 + (t^2/2!)y''_0 + (t^3/3!)y'''_0$, all derivatives at the start point.

Step 1 The DE is y' : $y' = e^t - 2y$

Step 2 Differentiate that. **Key point:** y is a function of t , so $d/dt(y) = y'$, not zero.

$$y'' = d/dt(e^t) - 2 \cdot d/dt(y) = e^t - 2y'$$

Step 3 Again: $y''' = e^t - 2y''$

Step 4 Now put in $t = 0$, $y = 1$ and work *downwards* — each line uses the one above.

$$y' = e^0 - 2(1) = 1 - 2 = -1$$

$$y'' = e^0 - 2(-1) = 1 + 2 = 3$$

$$y''' = e^0 - 2(3) = 1 - 6 = -5$$

Step 5 Substitute with $t = 0.1$ ($2! = 2$, $3! = 6$):

$$y = 1 + (0.1)(-1) + (0.01/2)(3) + (0.001/6)(-5)$$

$$= 1 - 0.1 + 0.015 - 0.000833$$

Answer $y(0.1) = 0.914167$

15 Newton's law of cooling: 300°F , room 80°F , 250°F after 10 min — find $T(20)$ 2x

NO CALCULUS NEEDED

Use the ready-made formula — you do not have to solve the differential equation:

$$T = T_0 + (T_1 - T_0)e^{-kt}, \quad T_0 = \text{room temp}, \quad T_1 = \text{starting temp.}$$

Step 1 $T_0 = 80$, $T_1 = 300$, so $T_1 - T_0 = 220$:

$$T = 80 + 220 e^{-kt}$$

Step 2 Use the 10-minute reading to pin down e^{-10k} :

$$250 = 80 + 220 e^{-10k} \rightarrow 170 = 220 e^{-10k} \rightarrow e^{-10k} = 170/220 = 17/22$$

Step 3 — the shortcut. You do **not** need k . Notice $e^{-20k} = (e^{-10k})^2$ — just square it:

$$e^{-20k} = (17/22)^2 = 289/484 = 0.597107$$

$$\text{Step 4} \quad T(20) = 80 + 220(0.597107) = 80 + 131.364$$

Answer $T(20) = 211.36^\circ\text{F}$

Coffee variant 92°C , room 24°C , 80°C after 1 min, when does it hit 65°C ?

$$T = 24 + 68e^{-kt}; \text{ at } t=1: 56 = 68e^{-k} \rightarrow e^{-k} = 14/17 \rightarrow k = \log(17/14) = 0.194156$$

$$\text{For } T=65: 41 = 68e^{-kt} \rightarrow e^{-kt} = 41/68 \rightarrow kt = \log(68/41)$$

$$t = \log(1.65854)/0.194156 = 0.505933/0.194156 = 2.606 \text{ minutes}$$

16 Orthogonal trajectory of cardioids $r = a(1-\cos\theta)$ 1x NEEDS CALCULUS

Given the family's DE is already handed to you: $dr/d\theta = r \cot(\theta/2)$.

Step 1 — the swap. For polar, replace $dr/d\theta$ by $-r^2(d\theta/dr)$:

$$-r^2(d\theta/dr) = r \cot(\theta/2)$$

Step 2 — tidy up. Divide both sides by r : $-r(d\theta/dr) = \cot(\theta/2)$.

Now separate — get all the r on one side, all the θ on the other:

$$-r d\theta = \cot(\theta/2) dr \rightarrow dr/r = -\tan(\theta/2) d\theta$$

Step 3 — integrate both sides.

$$\text{Left: } \int dr/r = \log r$$

$$\text{Right: } \int -\tan(\theta/2) d\theta. \text{ Using } \int \tan u du = -\log(\cos u) \text{ with } u = \theta/2 \text{ (so } d\theta = 2du \text{): this gives } +2 \log(\cos(\theta/2))$$

$$\text{So } \log r = 2 \log \cos(\theta/2) + \log c$$

Step 4 — undo the logs. $2 \log A = \log A^2$ and $\log A + \log B = \log AB$:

$$\log r = \log(c \cos^2(\theta/2)) \rightarrow r = c \cos^2(\theta/2)$$

Step 5 Use $\cos^2(\theta/2) = (1 + \cos\theta)/2$:

Answer $r = C(1 + \cos\theta)$ — the same cardioid, reflected.

17 If $r^n = k \cos(n\theta)$, find the DE of that family **1x** **NEEDS CALCULUS**

Aim Get rid of the parameter k . Taking logs first makes this easy.

Step 1 $\log(r^n) = \log k + \log(\cos n\theta) \rightarrow n \log r = \log k + \log \cos n\theta$

Step 2 Differentiate w.r.t. θ . $\log k$ is a constant $\rightarrow 0$.

Left: $n \cdot d/d\theta(\log r) = n \cdot (1/r)(dr/d\theta)$ (chain rule — r depends on θ)

Right: $d/d\theta(\log \cos n\theta) = (1/\cos n\theta) \cdot (-n \sin n\theta) = -n \tan n\theta$

Step 3 $n(1/r)(dr/d\theta) = -n \tan n\theta$. Cancel n :

Answer $(1/r)(dr/d\theta) = -\tan(n\theta)$, i.e. $dr/d\theta + r \tan(n\theta) = 0$

18 Orthogonal trajectory of $r = b \sin \theta$ **2x** **NEEDS CALCULUS**

Step 1 — eliminate b . $dr/d\theta = b \cos \theta$. Since $b = r/\sin \theta$:

$dr/d\theta = (r/\sin \theta) \cos \theta = r \cot \theta \rightarrow (1/r)(dr/d\theta) = \cot \theta$

Step 2 — swap. Replacing $dr/d\theta \rightarrow -r^2 d\theta/dr$ turns $\cot \theta$ into $-\tan \theta$:

$(1/r)(dr/d\theta) = -\tan \theta$

Step 3 — separate and integrate.

$dr/r = -\tan \theta d\theta \rightarrow \log r = -(-\log \cos \theta) + \log c = \log \cos \theta + \log c$

Answer $r = c \cos \theta$

UNIT III — Q5(b) or Q6(b)**21 Prove $(1/(D+a))X = e^{-ax} \int X e^{ax} dx$** **4x — the safest 4 marks in the paper** **light calculus**

What the notation means D means "differentiate", so $Dy = dy/dx$. $1/(D+a)$ means "undo $(D+a)$ ". Nothing to memorise numerically — this proof never changes.

Step 1 Call the answer y : $y = (1/(D+a))X$

Step 2 Multiply both sides by $(D+a)$ to clear the fraction:

$(D+a)y = X$, which written out is $dy/dx + ay = X$

Step 3 This is a standard linear equation $dy/dx + Py = Q$ with $P = a$.

Its integrating factor is $e^{\int a dx} = e^{ax} = e^{ax}$.

Step 4 Multiply the whole equation by e^{ax} :

$e^{ax}(dy/dx) + a e^{ax} y = X e^{ax}$

Step 5 — spot the product rule backwards. The left side is exactly what you get if you differentiate $y \cdot e^{ax}$:

$d/dx(y e^{ax}) = (dy/dx)e^{ax} + y(a e^{ax})$ ✓ same thing.

So the equation is simply $d/dx(y e^{ax}) = X e^{ax}$

Step 6 Integrate both sides w.r.t. x . On the left, integrating a derivative just undoes it:

$y e^{ax} = \int X e^{ax} dx$

Step 7 Divide by e^{ax} (dividing by e^{ax} is multiplying by e^{-ax}):

Answer $y = e^{-ax} \int X e^{ax} dx$ ■ Seven lines. Learn them.

22 General solution if the A.E. is $\lambda^3(\lambda+4)^2(\lambda^2+2\lambda+5)^2 = 0$ 1x NO CALCULUS

Step 1 — read off the roots.

$$\lambda^3 = 0 \rightarrow \lambda = 0, \text{ three times}$$

$$(\lambda+4)^2 = 0 \rightarrow \lambda = -4, \text{ twice}$$

$$(\lambda^2+2\lambda+5)^2 = 0 \rightarrow \text{solve } \lambda^2+2\lambda+5=0 \text{ by the quadratic formula:}$$

$$\lambda = [-2 \pm \sqrt{4-20}]/2 = [-2 \pm \sqrt{-16}]/2 = (-2 \pm 4i)/2 = -1 \pm 2i, \text{ each twice}$$

Step 2 — apply the rules.

• root m repeated k times $\rightarrow (C_1 + C_2x + \dots + C_k x^{k-1})e^{mx}$

• complex $\alpha \pm i\beta \rightarrow e^{\alpha x}(A \cos \beta x + B \sin \beta x)$; if repeated, the constants become linear in x

Step 3 — build each piece.

$$\lambda=0 \text{ three times} \rightarrow (C_1+C_2x+C_3x^2)e^{0x} = C_1+C_2x+C_3x^2 \text{ (since } e^0 = 1 \text{)}$$

$$\lambda=-4 \text{ twice} \rightarrow (C_4+C_5x)e^{-4x}$$

$$-1 \pm 2i \text{ twice} \rightarrow e^{-x}[(C_6+C_7x)\cos 2x + (C_8+C_9x)\sin 2x]$$

Answer $y = (C_1+C_2x+C_3x^2) + (C_4+C_5x)e^{-4x} + e^{-x}[(C_6+C_7x)\cos 2x + (C_8+C_9x)\sin 2x]$

23 Show $y^{iv} - k^4y = 0$ gives $C_1\cos kx + C_2\sin kx + C_3\cosh kx + C_4\sinh kx$ 1x NO CALCULUS

Step 1 Auxiliary equation: replace y^{iv} by m^4 , y by 1:

$$m^4 - k^4 = 0$$

Step 2 Factorise as a difference of two squares (twice):

$$m^4 - k^4 = (m^2)^2 - (k^2)^2 = (m^2 - k^2)(m^2 + k^2) = 0$$

Step 3 $m^2 = k^2 \rightarrow m = \pm k$ (real). $m^2 = -k^2 \rightarrow m = \pm ik$ (complex, $\alpha=0$, $\beta=k$).

Step 4 Write the C.F.:

$$y = Ae^{kx} + Be^{-kx} + e^{0x}(C \cos kx + D \sin kx)$$

Step 5 — convert the exponentials. By definition $\cosh kx = (e^{kx}+e^{-kx})/2$ and $\sinh kx = (e^{kx}-e^{-kx})/2$.

So $e^{kx} = \cosh kx + \sinh kx$ and $e^{-kx} = \cosh kx - \sinh kx$. Then

$$Ae^{kx} + Be^{-kx} = (A+B)\cosh kx + (A-B)\sinh kx. \text{ Rename } A+B = C_3, A-B = C_4.$$

Answer $y = C_1\cos kx + C_2\sin kx + C_3\cosh kx + C_4\sinh kx$ ■

24 Solve the BVP $y'' + 4y' + 4y = 0$, $y(1)=0$, $y'(0)=-1$ 1x light calculus

Step 1 A.E.: $m^2 + 4m + 4 = (m+2)^2 = 0 \rightarrow m = -2$, repeated.

Repeated root $\rightarrow y = (A + Bx)e^{-2x}$

Step 2 — differentiate, using the product rule. Let $u = A+Bx$, $v = e^{-2x}$.

$$u' = B, v' = -2e^{-2x}$$

$$y' = u'v + uv' = Be^{-2x} + (A+Bx)(-2e^{-2x}) = e^{-2x}[B - 2A - 2Bx]$$

Step 3 Apply $y'(0) = -1$. At $x=0$, $e^0 = 1$:

$$B - 2A = -1 \dots(i)$$

Step 4 Apply $y(1) = 0$:

$$(A + B)e^{-2} = 0. \text{ Since } e^{-2} \neq 0, \text{ we need } A + B = 0, \text{ so } B = -A \dots(ii)$$

Step 5 Put (ii) into (i): $-A - 2A = -1 \rightarrow -3A = -1 \rightarrow A = 1/3$, $B = -1/3$

Answer $y = (1 - x)e^{-2x} / 3$

25 Solve the IVP $y'' - y = 0$, $y(0)=3$, $y'(0)=3$ 1x almost no calculus

Step 1 A.E.: $m^2 - 1 = 0 \rightarrow m = +1$ and $m = -1 \rightarrow y = Ae^x + Be^{-x}$

Step 2 $y' = Ae^x - Be^{-x}$ (e^{-x} differentiates to $-e^{-x}$)

Step 3 At $x=0$, both exponentials are 1:

$$y(0) = A + B = 3 \dots(i) \quad y'(0) = A - B = 3 \dots(ii)$$

Step 4 Add (i) and (ii): $2A = 6 \rightarrow A = 3$. Then $B = 0$.

Answer $y = 3e^x$

26 Find the P.I. of $y'' - y' = x^2 + 2x + 4$ 1x NEEDS CALCULUS

Step 1 In operator form: $(D^2 - D)y = x^2 + 2x + 4$, so $y_p = (1/(D^2 - D))(x^2 + 2x + 4)$

Step 2 Factor out $-D$ so what is left starts with 1:

$$D^2 - D = -D(1 - D), \text{ so } y_p = -(1/D) \cdot (1/(1-D)) \cdot (x^2 + 2x + 4)$$

Step 3 Expand $1/(1-D)$ as a series (this is the binomial expansion $(1-D)^{-1} = 1 + D + D^2 + D^3 + \dots$). Stop at D^2 — higher terms kill a quadratic.

Step 4 Apply it. **Here D just means "differentiate".**

$$(1)(x^2 + 2x + 4) = x^2 + 2x + 4$$

$$(D)(x^2 + 2x + 4) = 2x + 2$$

$$(D^2)(x^2 + 2x + 4) = 2$$

$$\text{Add them: } x^2 + 2x + 4 + 2x + 2 + 2 = x^2 + 4x + 8$$

Step 5 Now the $-(1/D)$. $1/D$ means **integrate**, and the minus flips the sign:

$$\int (x^2 + 4x + 8) dx = x^3/3 + 2x^2 + 8x$$

Answer $y_p = -(x^3/3 + 2x^2 + 8x)$

27 Solve the Legendre equation $(2x+3)^2 y'' + 2(2x+3)y' - 12y = 0$ 1x NO CALCULUS after the swap

Step 1 — the substitution. Put $2x+3 = e^z$. Here $a = 2$ (whatever multiplies x).

The standard replacements are:

$$(2x+3)y' = 2Dy \quad (2x+3)^2 y'' = 2^2 D(D-1)y = 4D(D-1)y$$

The 4 is the thing everyone forgets.

Step 2 — substitute.

$$4D(D-1)y + 2(2Dy) - 12y = 0$$

$$4D^2y - 4Dy + 4Dy - 12y = 0$$

$$\text{The } -4Dy \text{ and } +4Dy \text{ cancel: } 4D^2y - 12y = 0$$

Step 3 Divide by 4: $D^2y = 3y$. A.E.: $m^2 = 3 \rightarrow m = \pm\sqrt{3}$

Step 4 $y = C_1 e^{\sqrt{3}z} + C_2 e^{-\sqrt{3}z}$

Step 5 — substitute back. Since $e^z = 2x+3$, we get $e^{\sqrt{3}z} = (e^z)^{\sqrt{3}} = (2x+3)^{\sqrt{3}}$.

Answer $y = C_1(2x+3)^{\sqrt{3}} + C_2(2x+3)^{-\sqrt{3}}$

28-29 Writing the C.F. from given roots 2x NO CALCULUS

Roots $-3 \pm i$, 2 , 2 , 2 :

$$-3 \pm i \text{ is complex with } \alpha = -3, \beta = 1 \rightarrow e^{-3x}(C_1 \cos x + C_2 \sin x)$$

$$2 \text{ repeated three times } \rightarrow (C_3 + C_4 x + C_5 x^2)e^{2x}$$

$$y = e^{-3x}(C_1 \cos x + C_2 \sin x) + (C_3 + C_4 x + C_5 x^2)e^{2x}$$

4th-order, two roots imaginary and two real distinct:

Call them $\alpha \pm i\beta$ and m_1, m_2 :

$$y = e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x) + C_3 e^{m_1 x} + C_4 e^{m_2 x}$$

UNIT IV — Q7(b) or Q8(b) — almost all pure arithmetic

Start your revision here. Every question in this unit is building a table of differences and substituting into a formula. **There is no calculus at all.**

31 Missing terms: $x = 45, 50, 55, 60, 65$; $y = 3.0, -, 2.0, -, -2.4$ 2x NO CALCULUS

The idea You have **3 known** values. Three points determine a *quadratic*. For a quadratic, all **third** differences are zero. That gives two equations for the two unknowns.

Step 1 Call the unknowns a and b : $y = 3.0, a, 2.0, b, -2.4$

Step 2 Build the difference table. Each entry = (the one below) – (the one above).

Y	1ST DIFFERENCE	2ND DIFFERENCE	3RD DIFFERENCE
3.0	$a - 3$		
a	$2 - a$	$(2-a)-(a-3) = 5 - 2a$	
2.0	$b - 2$	$(b-2)-(2-a) = a + b - 4$	$(a+b-4)-(5-2a) = 3a + b - 9$
b	$-2.4 - b$	$(-2.4-b)-(b-2) = -0.4 - 2b$	$(-0.4-2b)-(a+b-4) = 3.6 - a - 3b$
-2.4			

Step 3 Set **both** third differences to zero:

$$3a + b - 9 = 0 \quad \dots(i) \quad 3.6 - a - 3b = 0 \quad \dots(ii)$$

Step 4 From (i): $b = 9 - 3a$. Put into (ii):

$$3.6 - a - 3(9 - 3a) = 0 \rightarrow 3.6 - a - 27 + 9a = 0 \rightarrow 8a = 23.4 \rightarrow a = 2.925$$

$$b = 9 - 3(2.925) = 9 - 8.775 = 0.225$$

Answer $y(50) = 2.925, y(60) = 0.225$

Step 5 — check (do this, it earns a mark). Fill them in: $3.0, 2.925, 2.0, 0.225, -2.4$

1st: $-0.075, -0.925, -1.775, -2.625$ 2nd: $-0.85, -0.85, -0.85$ ✓ constant.

Warning One equation is not enough. Setting a single *fourth* difference to zero gives only $a + b = 3.15$, which you cannot solve. You need **both third differences**.

32 Missing terms: $x = 1, \dots, 7$; $y = 2, 5, -, 13, 22, -, 63$ 1x NO CALCULUS

Different question — count the knowns. Here there are **5 known** values, so a *quartic* (degree 4) fits, and it is the **fifth** differences that vanish.

Easier route Fit the quartic straight through the five known points $(1, 2), (2, 5), (4, 13), (5, 22), (7, 63)$ using divided differences, then read off $x=3$ and $x=6$.

$$P(x) = x^4/60 + 2x^3/15 - 71x^2/60 + 161x/30 - 7/3$$

Answer $y(3) = 121/15 = 8.067, y(6) = 113/3 = 37.667$

Not round numbers — that is correct here, do not "tidy" them.

33-34 Trapezoidal rule 2x NO CALCULUS

The rule $\int_y dx = (h/2)[(\text{first} + \text{last}) + 2 \times (\text{everything in between})]$, $h = (b-a)/n$

(a) $\int_0^1 dx/(1+x^2)$, 4 strips. $h = 1/4 = 0.25$.

Compute $y = 1/(1+x^2)$ at each x :

$$x=0: 1/1 = 1.000000 \quad x=0.25: 1/1.0625 = 0.941176 \quad x=0.5: 1/1.25 = 0.800000$$

$$x=0.75: 1/1.5625 = 0.640000 \quad x=1: 1/2 = 0.500000$$

$$I = (0.25/2)[(1 + 0.5) + 2(0.941176 + 0.8 + 0.64)]$$

$$= 0.125[1.5 + 2(2.381176)] = 0.125[1.5 + 4.762352] = 0.125(6.262352)$$

$$I = 0.782794 \quad (\text{exact } \pi/4 = 0.785398)$$

(b) $\int_0^1 e^x dx$, steps of 0.2. $h = 0.2$, five strips.

$$y = 1, 1.221403, 1.491825, 1.822119, 2.225541, 2.718282$$

$$I = 0.1[(1 + 2.718282) + 2(1.221403 + 1.491825 + 1.822119 + 2.225541)]$$

$$= 0.1[3.718282 + 2(6.760888)] = 0.1[3.718282 + 13.521776]$$

$$I = 1.724006 \quad (\text{exact } e - 1 = 1.718282)$$

35 Backward difference table for $y = 2x^2 - x + \cos 2$ on $(2,3)$, $h=0.2$ **1x****NO CALCULUS**

Note $\cos 2$ is a *constant* (2 radians), $= -0.416147$. Not $\cos(2x)$.

Step 1 Compute y at $x = 2, 2.2, 2.4, 2.6, 2.8, 3.0$:

$x=2$: $8 - 2 - 0.416147 = 5.583853$ · $x=2.2$: $9.68 - 2.2 - 0.416147 = 7.063853$
then $8.703853, 10.503853, 12.463853, 14.583853$

Step 2 Differences:

∇ : $1.48, 1.64, 1.80, 1.96, 2.12$

∇^2 : $0.16, 0.16, 0.16, 0.16$ — all equal, as they must be for a quadratic.

Answer $\nabla^2 y_3 = 0.16$ *Sanity check: for ax^2 the 2nd difference is $2ah^2 = 2(2)(0.04) = 0.16$ ✓*

36 Estimate $\cosh(0.56)$ from a table**1x****NO CALCULUS**

Data $x = 0.5, 0.6, 0.7, 0.8$; $y = 1.1276, 1.1855, 1.2552, 1.3374$. $h = 0.1$.

Step 1 0.56 is near the *start* → use Newton **forward**.

Δ : $0.0579, 0.0697, 0.0822$ · Δ^2 : $0.0118, 0.0125$ · Δ^3 : 0.0007

Step 2 $p = (x - x_0)/h = (0.56 - 0.5)/0.1 = 0.6$

Step 3 $y = y_0 + p\Delta y_0 + p(p-1)/2 \cdot \Delta^2 y_0 + p(p-1)(p-2)/6 \cdot \Delta^3 y_0$

$= 1.1276 + 0.6(0.0579) + (0.6)(-0.4)/2 \times 0.0118 + (0.6)(-0.4)(-1.4)/6 \times 0.0007$

$= 1.1276 + 0.03474 - 0.001416 + 0.0000392$

Answer $\cosh(0.56) \approx 1.16096$ (true 1.16094)

37-38 Lagrange interpolation, forward and inverse**2x****NO CALCULUS**

(a) Find $f(1)$ from $(-1,9)$, $(0,5)$, $(2,3)$.

Each term is y_i times a fraction: on top, $(x - \text{the other two } x\text{'s})$; on the bottom, $(x_i - \text{the other two})$.

$P(x) = 9 \cdot (x-0)(x-2)/((-1-0)(-1-2)) + 5 \cdot (x+1)(x-2)/((0+1)(0-2)) + 3 \cdot (x+1)(x-0)/((2+1)(2-0))$

Denominators: $(-1)(-3) = 3$, $(1)(-2) = -2$, $(3)(2) = 6$

Put $x = 1$: $9(1)(-1)/3 + 5(2)(-1)/(-2) + 3(2)(1)/6 = -3 + 5 + 1$

$f(1) = 3$ (the full polynomial is $x^2 - 3x + 5$)

(b) Inverse: find x when $y=15$ from $(6,13)$, $(9,14)$, $(11,16)$.

Identical formula with x and y swapped:

$x = 6 \cdot (15-14)(15-16)/((13-14)(13-16)) + 9 \cdot (15-13)(15-16)/((14-13)(14-16)) + 11 \cdot (15-13)(15-14)/((16-13)(16-14))$

$= 6(1)(-1)/((-1)(-3)) + 9(2)(-1)/((1)(-2)) + 11(2)(1)/((3)(2))$

$= 6(-1/3) + 9(1) + 11(1/3) = -2 + 9 + 3.667$

$x = 32/3 = 10.667$

UNIT V — Q9(b) or Q10(b) — entirely arithmetic

41-42 Gauss-Seidel 3x NO CALCULUS

System A $2x+17y+4z=35$; $3x+y+10z=24$; $28x+4y-z=32$, start $(0,0,0)$.

Step 1 — REORDER first. Each row needs its diagonal entry to be the biggest in that row.

Row with the big x is $28x+4y-z=32$ → put it **first**.

Row with the big y is $2x+17y+4z=35$ → **second**.

Row with the big z is $3x+y+10z=24$ → **third**.

Step 2 — make each variable the subject.

$$x = (32 - 4y + z)/28 \quad y = (35 - 2x - 4z)/17 \quad z = (24 - 3x - y)/10$$

Step 3 — iterate, always using the newest values.

Iteration 1 (start $y=0$, $z=0$):

$$x = (32 - 0 + 0)/28 = 1.142857$$

$$y = (35 - 2(1.142857) - 0)/17 = (35 - 2.285714)/17 = 1.924370 \quad \leftarrow \text{uses the new } x$$

$$z = (24 - 3(1.142857) - 1.924370)/10 = (24 - 3.428571 - 1.924370)/10 = 1.864706$$

Iteration 2:

$$x = (32 - 4(1.924370) + 1.864706)/28 = 0.934544$$

$$y = (35 - 2(0.934544) - 4(1.864706))/17 = 1.510123$$

$$z = (24 - 3(0.934544) - 1.510123)/10 = 1.968625$$

Answer **After 2 iterations: $x = 0.934544$, $y = 1.510123$, $z = 1.968625$** (heading for 1.001, 1.482, 1.952)

System B $5x+2y+z=12$; $x+4y+2z=15$; $x+2y+5z=20$, start $(1,0,0)$. Already dominant — no reorder.

$$x=(12-2y-z)/5, \quad y=(15-x-2z)/4, \quad z=(20-x-2y)/5$$

It1: $x = 12/5 = 2.4$; $y = (15-2.4-0)/4 = 3.15$; $z = (20-2.4-6.3)/5 = 2.26$

It2: $x = (12-6.3-2.26)/5 = 0.688$; $y = (15-0.688-4.52)/4 = 2.448$; $z = (20-0.688-4.896)/5 = 2.8832$

$x = 0.688$, $y = 2.448$, $z = 2.8832$ (exact answer is 1, 2, 3)

43 Rayleigh's power method, 2 iterations 1x NO CALCULUS

Given $A = \begin{bmatrix} 2 & -2 & 1 \\ 1 & 3 & -1 \\ -1 & 2 & 2 \end{bmatrix}$, $X_0 = [1, 0, 0]^T$.

The method Multiply A by the vector; pull out the largest-size number as λ ; what remains is the next vector.

Iteration 1 AX_0 = each row of A dotted with $[1, 0, 0]$ = just the first column:

$[2, 1, -1]^T$. Largest size is 2, so factor it out:

$$= 2[1, 0.5, -0.5]^T \rightarrow \lambda_1 = 2, \quad X_1 = [1, 0.5, -0.5]^T$$

Iteration 2 AX_1 :

$$\text{Row 1: } 2(1) + (-2)(0.5) + 1(-0.5) = 2 - 1 - 0.5 = 0.5$$

$$\text{Row 2: } 1(1) + 3(0.5) + (-1)(-0.5) = 1 + 1.5 + 0.5 = 3$$

$$\text{Row 3: } -1(1) + 2(0.5) + 2(-0.5) = -1 + 1 - 1 = -1$$

$$= [0.5, 3, -1]^T = 3[0.1667, 1, -0.3333]^T$$

Answer **$\lambda = 3$, $X = [0.167, 1, -0.333]^T$**

Note This particular matrix has *complex* eigenvalues, so the method never really settles. The question asks for two iterations — do them and stop. If the numbers look unstable, that is the matrix, not you.

44 Find x so the matrix has rank 2 1x NO CALCULUS

Idea A 3×3 matrix has rank 3 unless its determinant is zero. So set $|A| = 0$.

Given $A = \begin{bmatrix} 4-x & 2\sqrt{5} & 0 \\ 2\sqrt{5} & 4-x & \sqrt{5} \\ 0 & \sqrt{5} & 4-x \end{bmatrix}$. Write $u = 4-x$.

Step 1 Expand along the top row:

$$|A| = u(u \cdot u - \sqrt{5} \cdot \sqrt{5}) - 2\sqrt{5}(2\sqrt{5} \cdot u - \sqrt{5} \cdot 0) + 0$$

$$= u(u^2 - 5) - 2\sqrt{5}(2\sqrt{5}u) = u^3 - 5u - 20u = u^3 - 25u = u(u^2 - 25)$$

Step 2 Set to zero: $u = 0$ or $u = \pm 5$, i.e. $4-x = 0, 5, -5$

Answer **$x = 4$, $x = -1$, $x = 9$** (at each of these a 2×2 minor is still non-zero, so the rank is exactly 2)

45 Prove: if λ is an eigenvalue of A then $1/\lambda$ is one of A^{-1} 1x NO CALCULUS

Step 1 By definition, $AX = \lambda X$ for some non-zero X .

Step 2 Multiply both sides *on the left* by A^{-1} :

$$A^{-1}(AX) = A^{-1}(\lambda X)$$

Step 3 On the left, $A^{-1}A = I$ and $IX = X$. On the right, λ is just a number so it slides out front:

$$X = \lambda(A^{-1}X)$$

Step 4 Divide by λ . (Allowed: A is invertible, so no eigenvalue is 0.)

Answer $A^{-1}X = (1/\lambda)X$, which says exactly that $1/\lambda$ is an eigenvalue of A^{-1} . ■

If calculus is the problem, revise in this order

1. Unit V (Gauss-Seidel, Rayleigh, rank, the eigenvalue proof) — no calculus at all.

2. Unit IV (difference tables, missing terms, Lagrange, trapezoidal) — no calculus at all.

3. Unit II's numerical half (Euler, bungee jumper) — arithmetic once you have $f(x,y)$.

4. Unit III's $1/(D+a)$ proof — seven lines, appeared **4x**, and you now have every one of them.

5. Unit III's C.F. questions — solving a quadratic and writing down a pattern.

Only after all that: the expansions and orthogonal trajectories, which genuinely need differentiation and integration.

That order alone is 4 of the 5 four-markers — 16 of the 20 marks — with barely any calculus.

Every derivative and integral used is listed in the toolkit at the top; nothing below it assumes prior calculus. All numerical results computed and verified.

nx = appearances across the 6 papers.